



Fiona Stanley Fremantle Hospitals Group

Policy and Procedure

Alcohol and Other Drugs: management of addiction, toxicity and withdrawal

Reference #: FSFH-HW-POL-0063

Scope

Site	Service/Department/Unit	Disciplines
Fiona Stanley Hospital Fremantle Hospital	Hospital Wide	Medical, Nursing, Midwifery, Pharmacy, Allied Health

1. Introduction

Early detection of withdrawal and the prevention of risks associated with withdrawal are key components of effective and safe patient management. This policy and procedure will establish the minimum practice standards for the care of patients who may be affected by alcohol or other drugs. Information is provided specific to alcohol or drug group including initial considerations for patient care, signs and symptoms of intoxication and withdrawal, assessment and management.

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The information provided in this document is based on its relevance to Fiona Stanley Fremantle Hospital Group (FSFHG) only. Due to differences in context, scope of practice and differences in service delivery, FSFHG does not make claim to its relevance or appropriateness for use within other organisations/sites.

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3. Terminology

Alcohol	Alcohol is a central nervous system depressant/sedative drug. Even in small doses alcohol significantly impedes motor and cognitive function depending on the dose and individual susceptibility.
Drug	Any substance (except for food and water) which, when taken into the body, alters the body's function either physically and/or psychologically. Drugs may be legal e.g. alcohol, prescription medication or illicit e.g. methamphetamine.
Intoxication	A condition that follows the administration of a psychoactive substance and results in disturbances in the level of consciousness, cognition, judgement, affect, or behaviour, or other psychophysiological functions and responses.
Dependence syndrome	A cluster of behavioural, cognitive, and physiological phenomena that develop after repeated substance use and that typically include a strong desire to take the drug, difficulties in controlling its use, persisting in its use despite harmful consequences, a higher priority given to drug use than to other activities and obligations, increased tolerance, and sometimes a physical withdrawal state (ICD-10).
Withdrawal state	A group of symptoms of variable clustering and severity occurring on absolute or relative withdrawal of a psychoactive substance after persistent use of that substance (ICD-10). The onset and course of the withdrawal state are time-limited and are related to the type of psychoactive substance and dose being used immediately before cessation or reduction of use. The withdrawal state may be complicated by convulsions.
Pharmaceutical Benefits Scheme (PBS)	The Australian Government's PBS provides reliable, timely and affordable access to a wide range of medicines for all Australians.
Wernicke's Encephalopathy (WE)	An acute reversible neuropsychiatric condition which presents in alcohol dependent individuals due to acute thiamine deficiency. Acute WE is characterised by the classical triad of symptoms of cerebellar ataxia, ocular abnormalities and a state of confusion.
Korsakoff's	A largely irreversible condition characterised by short term

Syndrome (KS)	memory loss, decreased learning ability and compensatory confabulation, due to deficiency of thiamine. This is a progression of untreated or inadequately treated WE.
Women and Newborn Drug and Alcohol Service (WANDAS)	The state-wide tertiary service that provides specialist clinical services for the care of pregnant women with drug and alcohol dependence. The service is provided by King Edward Memorial Hospital, contact WANDAS
Drug and Alcohol Clinical Advisory Service (DACAS)	A specialised telephone consultancy service that provides clinical advice to health professionals across WA on all issues relating to management of patients with alcohol and other drug use. This service is available 08:00 to 20:00 hrs Monday to Friday. Phone (08) 6553 0520.
Pro re nata (PRN)	Term used to indicate when medicines are to be used 'as required'
Quarter in die (QID)	Indicating four times daily or six hourly.
Urine drug screen (UDS)	A rapid, visual panel of immunoassays that can be used for the simultaneous qualitative detection of multiple drug metabolites in human urine. Widely accepted method for alcohol and other drug screening.
Clinical Institute Withdrawal Assessment for Alcohol (CIWA-Ar)	A proven, valid and sensitive instrument for assessing the clinical course of simple alcohol withdrawal. It assists clinicians to assess the severity of symptoms and timing of assessments to guide treatment appropriately.
Alcohol Smoking Substance Involvement Screening Tool (ASSIST-Lite)	A short risk identification tool for use with adults (aged 18 or over) covering alcohol, tobacco, cannabis, stimulants, sedatives, opioids and other psychoactive substances including the use of medicines not as prescribed. The ASSIST-Lite allows for assessment of risk and formulation of action/management plans.
Delirium tremens (DT)	Is defined by hallucinations, disorientation, tachycardia, hypertension, hyperthermia, agitation and diaphoresis in the setting of acute reduction or abstinence from alcohol. Approximately 5% of patients who undergo withdrawal from alcohol suffer from DT.

4. Policy

- Alcohol and other drugs (AOD) usage screening should occur in pre-admission clinics or during assessment in the Emergency Department (ED), where possible.

- Many individuals who use AOD have co-occurring mental health issues. Patients presenting with either AOD issues or mental health issues should be assessed and treated concurrently¹.
- Withdrawal management may only be the first stage of alcohol and other drug treatment and by itself does little to change long-term drug dependence. Patients will be encouraged to continue drug treatment following withdrawal¹.
- All patients at risk of experiencing alcohol withdrawal are to be commenced on an Alcohol Withdrawal Chart (CIWA-Ar)

It is mandatory that all patients experiencing acute alcohol withdrawal are provided with thiamine replacement therapy. Refer to Section 6.7.

- Oral thiamine is ineffective in moderate to severe thiamine deficiency and is not to be used in the ED.

Intravenous (IV) glucose should not be given before thiamine administration as it may precipitate, WE in alcohol dependent patients. Glucose supplements if required may be given after thiamine administration.

- All patients are to be advised on admission that the use of alcohol and/or illicit drugs will not be permitted whilst they are in hospital.
- If there is reasonable concern that a patient has consumed AOD as an inpatient they may be requested to submit to breath, urine and/or blood alcohol screening.
- All breathalyser testing must be performed by appropriately trained and experienced clinicians with relevant medical staff consulted in the decision to test the patient.
- The Medicines and Poisons Act 2014 requires authorised medical practitioners / nurse practitioners to report to the Department of Health (DoH) Medicines and Poisons Regulation Branch within 48 hours, when they have reason to believe a person has acquired, as a result of repeated administration of drugs of addiction, an overwhelming desire for the continued administration of a drug of addiction. Reports of dependence will be made using the approved 'Reporting a Drug Dependent Person' (refer to section 9.8) or 'Reporting an Oversupplied Person' form.
- Prior written authorisation must be received from the DoH before prescribing Schedule 8 discharge medicines to a person who has been recorded as a Drug Dependent Person.
- The Health (Notification of Stimulant Induced Psychosis) Regulations 2010 requires a psychiatrist to notify the DoH Medicines and Poisons Regulation Branch within 72 hours of making a diagnosis of stimulant induced psychosis. This is made by completing the 'Notification Stimulant Induced Psychosis' form.

5. Screening and assessment

5.1. Initial considerations

- The patient's mental health and current form of thought, perception and behaviour may be affected by recent substance use e.g. intoxication/withdrawal, psychosis, impaired cognitive function, suicidal/homicidal ideation.
- Longer periods of use may complicate treatment and compromise recovery goals.
- The patient's AOD use may impact their legal competency in relation to consent to treatment.
- Issues with AOD can affect anyone. Some groups in the community are more at risk than others:
 - People with mental health conditions
 - People with a history of trauma who may be seeking 'emotional blockade'
 - Pregnant people
 - People with poly substance use
 - People living in remote and very remote areas²
 - Aboriginal & Torres Strait Islander Peoples who may have:
 - been impacted by government policies³
 - trauma in its many forms from past policies causing post-traumatic stress
 - intergenerational trauma which may result in substance misuse
 - limited access to culturally appropriate care and treatment
 - Lesbian, gay, bisexual, transgender, intersex (LGBTIQ+) communities⁴
 - Young people or older adults
 - People experiencing domestic abuse
 - People with acquired brain injury or cognitive impairment
 - Homeless people or people with housing instability
 - Culturally and linguistically diverse people
- Consider referral to:
 - appropriate Liaison Services
 - FSFHG Interpreter Booking policy.
- The WANDAS may be contacted for specialist advice for pregnant patients.

5.2. AOD screening

- A simple routine AOD screen occurs for all patients seen in any setting excluding patients with altered conscious state or acute clinical condition.

- Where a screening tool has not been completed on admission this must be handed over to the next shift to complete.
- Screening tools:
 - The FSH AOD Service use the Alcohol Smoking Substance Involvement Screening Tool (ASSIST-Lite) to initially screen all patients presenting to the ED, general wards and outpatient settings.
 - Clinicians in the FSH Alcohol and Other Drug Outpatient Clinic will use the Australian Treatment Outcomes Profile (ATOP) to monitor substance use levels over four weeks.
 - The FH Dual Diagnosis Service (Mental Health Service) uses the ASSIST-Lite to assess risk and formulate appropriate action/management plans.
- Initial screening identifies those patients at risk of withdrawal and whether monitoring using a CIWA-Ar or Clinical Opioid Withdrawal scale (COWS) is clinically indicated.
- When indicated, the patient and/or carer will be provided with an AOD brief intervention, ASSIST-Lite feedback pamphlet and a [Mental Health Commission Self Help Guide booklet](#).
- A standard drinks guide is provided to assist in AOD screening. Refer to [Appendix 1](#)

The CIWA-AR should be commenced where any predictors of alcohol withdrawal are identified.

The COWS should be commenced where any predictors of opioid withdrawal are identified.

5.3. AOD assessment

- AOD assessments should be biopsychosocial not just medical. Issues that are related to and can exacerbate and prolong withdrawal include:
 - mental health
 - legal
 - parenting/family/child protection
 - housing
 - domestic violence
- Patient history taking will include:
 - blood borne viruses (BBV)
 - sexually transmitted infections (STI)
 - pregnancy

- seizure history
- physical co-morbidities
- mental health co-morbidities
- social history including dependents, housing
- cognitive assessment.
- Physical examination will include:
 - general/focussed physical examination
 - evidence of intoxication
 - evidence of withdrawal
 - evidence of complications from substance use.
- Laboratory and other testing will include:
 - full blood count
 - serum electrolytes including potassium, magnesium and phosphate
 - glucose
 - creatinine
 - liver function tests
 - lipase
 - blood alcohol level
 - urine drug testing
 - beta human chorionic gonadotropin (HCG).

5.4. Referral to specialist services

- Referrals to specialist FSFHG AOD Consultation Liaison Services will be initiated when clinically indicated.
- For FSH patients:
 - ED patient can be referred through the Emergency Department Information System (EDIS), by phone or direct consult
 - Inpatient wards referrals are via e-Referral and phone call
 - Outpatient areas utilise the [Mental Health Commission](#) website for resources or contact CAS directly
- For FH Mental Health Service refer to the Dual Diagnosis Nurse:
 - via e-Referral for inpatients
 - by completing a Mental Health Services referral form for outpatients.
- For FH general wards and outpatients:
 - contact the Mental Health Consultation Liaison Service or CAS.

6. Alcohol

6.1. Intoxication signs and symptoms

- Intoxication can be exacerbated by the intake of other medications. Refer to [Appendix 2](#) Drugs with the potential to interact with alcohol.
- Alcohol intoxication may present with:
 - varying degrees of talkativeness, euphoria, disinhibited and bizarre behaviour
 - ataxia, uncoordinated motor function
 - pressured or slurred speech
 - double vision, impaired memory, impaired judgement
 - hypothermia, hypotension
 - nausea, vomiting abdominal pain
 - dysrhythmia, seizures
 - sleepiness, poor responses to external stimuli
 - dual diagnosis presentations including anxiety, depression, elevated mood, hallucinations, delusions.
 - respiratory depression, coma, death.

6.2. Clinical management

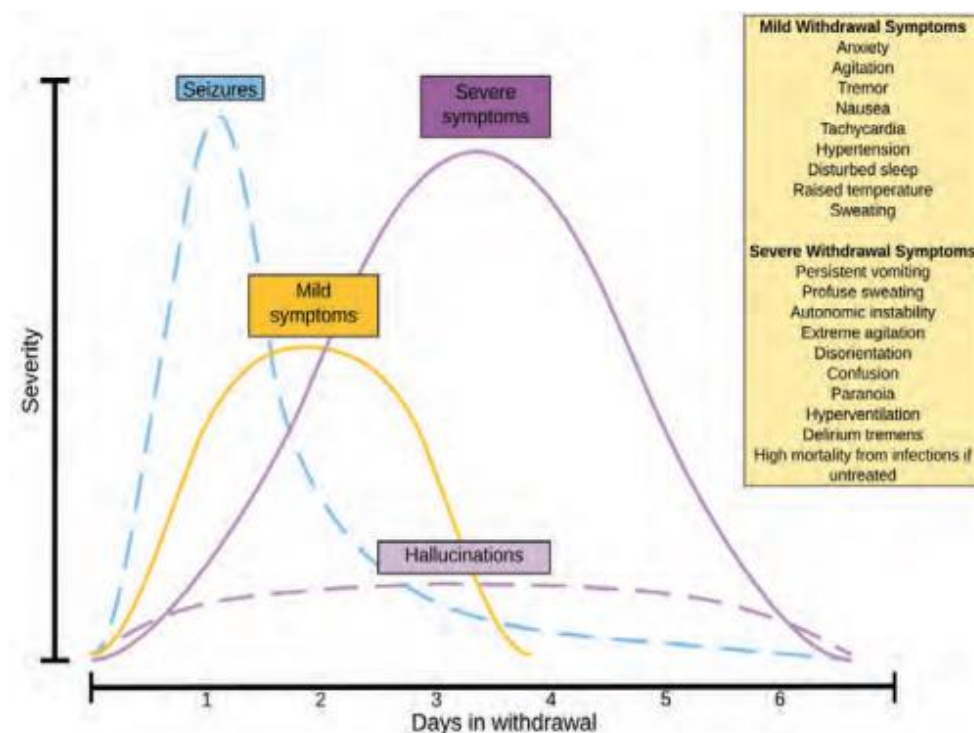
- Clinical management will be based on the clinical condition of the patient and the escalation protocols on the appropriate Observation and Response Chart and CIWA-Ar and in consultation with the Medical Officer (MO)/Senior Clinician and the AOD service
- The level of intoxication will be assessed by breath alcohol reading or blood alcohol concentration test.
 - Alcohol blood levels reduce by 0.015 – 0.02 % per hour on average.
- Assess mental state and discuss any concerns with the MO/Senior Clinician.
- Where possible reduce environmental stimuli e.g. provide single room, dim lights.
- Medical review must occur if there is any identified change to the patient's neurological observations and assessments including Glasgow Coma Scale, speech, eye movements and pupil abnormalities.
- Monitor urine output if there is evidence of potential renal impairment.
- Refer to Section 12 for general nursing care requirements.

6.3. Alcohol withdrawal

- Initial considerations:
 - The signs and symptoms of alcohol withdrawal may be confused with other factors that may be present in critical care patients such as electrolyte imbalances, pain, and infection.
 - The risk of major withdrawal is greater in people with acute medical conditions (e.g. trauma or sepsis).
 - Onset of alcohol withdrawal is usually 6-24 hours after the last drink and can occur before alcohol level is <0.01% or breath alcohol is 0.000%.
 - In people with severe alcohol dependence withdrawal can occur when the blood alcohol level is decreasing, even if the patient is still intoxicated.
 - Seizures affect 5% of patients, occurring early usually 7-24 hours after the last drink.
 - Seizures are tonic clonic in type i.e. generalised not focal and usually (though not always) occur as a single seizure.

Table 1 Signs and symptoms of alcohol withdrawal classification

Withdrawal	Autonomic hyperactivity	Gastrointestinal features	Cognitive and perceptual changes
Uncomplicated	Sweating Tachycardia Hypertension Tremor Fever (generally < 38°C)	Anorexia Nausea Vomiting Dyspepsia Diarrhoea	Poor concentration Anxiety Psychomotor agitation Disturbed sleep Vivid dreams
Severe & complex	Dehydration and electrolyte disturbance		Seizures Hallucinations or perceptual disturbances (visual, tactile, auditory) Delirium

Figure 1. Stages of alcohol withdrawal and symptoms⁵

(Source: NSW Department of Health, 2008)

Table 2. Predictors of alcohol withdrawal severity

Predictor	Severity
Current drinking patterns	- The severity of withdrawal is only moderately predicted by the amount of alcohol consumed. - Chronic heavy alcohol use (15 standard drinks or more per day) is associated with greater withdrawal severity than lower levels of use. - A predictor of increased alcohol withdrawal severity is the onset of withdrawal symptoms (e.g. tremor, nausea, anxiety) upon waking that are normally relieved by early morning drinking. - Individuals with heavy but irregular alcohol use (2-3 days per week), sometimes referred to as binge drinking, generally do not experience severe withdrawal.
Past withdrawal experience	Patients with a history of severe alcohol withdrawal syndrome (e.g. severe anxiety, seizure, delirium and hallucinations) are more likely to experience similar complications in future withdrawal episodes.

Poly substance use	• Patients with a heavy or regular use of other substances (e.g. benzodiazepines, stimulants, opioids) may experience more severe and complex withdrawal features. • Withdrawal from both alcohol and benzodiazepines may increase the risk of withdrawal complications. • Refer Appendix 2 Drugs with the potential to interact with alcohol.
Medical or psychiatric co morbidity	• Patients with concomitant medical conditions (e.g. sepsis, epilepsy, severe hepatic disease, head injury, pain, and nutritional depletion) or psychiatric conditions (e.g. anxiety, psychosis or depression) could be more likely to experience severe and complex withdrawal complications.

6.4. Acute alcohol delirium

- Acute alcohol withdrawal delirium can be difficult to distinguish from other forms of delirium and in the absence of a comprehensive history, alcohol withdrawal may go untreated. Refer to the FSFHG Cognitive Impairment Hub page.

Delirium tremens (DT) is a medical emergency and typically begins between 48 and 96 hours after the last drink and may last 1-5 days (or longer).

- DT is predicted during alcohol withdrawal by CIWA-Ar scores above 15 - especially in association with a systolic blood pressure >150mmHg, or a pulse rate > 100 beats per minute.
- Patients who develop DT will generally require intravenous (IV) fluids and IV sedation and should be managed in a high dependency unit or intensive care unit ^{6,7}.
- Alcoholic hallucinations develop within 12 to 24 hours of abstinence and typically resolve within 24 – 48 hours (which is the earliest point at which DT typically develops)^{6,7}.
- Predictors of alcohol withdrawal severity or complications may include^{6,7}
 - heavy recent alcohol use patterns – dose, regularity and duration of use
 - heavy or regular use of multiple substances
 - co-existing medical or psychiatric illness
 - past severe withdrawal experience
- Patients with any of these factors may be at a higher risk of severe or complicated withdrawal and may benefit from early AOD clinical staff involvement in assessment and withdrawal management.

6.5. Aims of alcohol withdrawal management

- Effective alcohol withdrawal management is through symptom identification and treatment. It aims to prevent:
 - Wernicke's encephalopathy
 - progression to severe withdrawal
 - patient falls
 - injury to self and others due to mental state
 - dehydration
 - electrolyte and nutritional disorders
 - potential for seizures
 - complications due to underlying illness

6.6. Pharmaceutical management

- Benzodiazepines are the treatment of choice for alcohol withdrawal due to good oral absorption, rapid onset and prolonged duration of effects⁸.
- Benzodiazepines should not be abruptly ceased but slowly reduced in a strict regime. Consider specialist advice re titrating regime and risk reduction strategies such as staged supply.

Use caution prescribing benzodiazepines to patients with respiratory failure, elderly patients, liver failure patients or those with a recent head injury. Consult with a senior doctor to ensure correct dosing.

- In patients with decompensated cirrhosis and alcohol use disorder lorazepam will be used instead of diazepam due to shorter acting duration and minimisation of accumulation. Refer to the Decompensated cirrhosis – Early Management Guideline.

1mg of lorazepam per oral is equivalent to 5mg of diazepam per oral

- The three most used benzodiazepine regimens are symptom triggered therapy, loading dose therapy and fixed schedule therapy. Refer to Table 3.
- The setting in which the withdrawal is occurring will be considered when deciding on the treatment regime.

Table 3. Pharmaceutical management of alcohol withdrawal

Treatment regimen	Drug dose	Setting	Patient
Loading dose therapy	Diazepam 10-20mg orally every 2 hours until 60 – 80mg given or lightly sedated. Maximum dose 120mg Diazepam in 24 hours. Reducing over 3-5 days with fixed dosing regimen plus CIWA-Ar	ED/Intensive care areas	Severe withdrawal CIWA-Ar >20 High risk of complex withdrawal e.g. delirium, seizure, hallucinations. History withdrawal seizures.
Fixed schedule therapy	Day 1: 20mg diazepam QID plus CIWA-Ar for additional prn if indicated. Reducing doses over 4-5 days	General wards Mental Health areas ED	Moderate to severe alcohol withdrawal in medical setting (CIWA-Ar > 9). History of significant withdrawal symptoms, withdrawing from alcohol and other substances. Significant medical or psychiatric co - morbidity.
Symptom triggered therapy	PRN Diazepam to be prescribed and given as per CIWA-Ar if required.	General wards Mental Health areas ED	Mild to moderate alcohol withdrawal (CIWA-Ar 0-8) No history of alcohol withdrawal complications. No significant co - morbidity

6.7. Thiamine replacement therapy

- Thiamine deficiency is common in patients with chronic misuse of alcohol and may lead to potentially life-threatening conditions such as WE.
- Thiamine replacement therapy reduces the risk of precipitating WE and/or Korsakoff's Syndrome.
- There is little evidence to support what is the optimal regime of thiamine supplementation to prevent neurotoxicity.

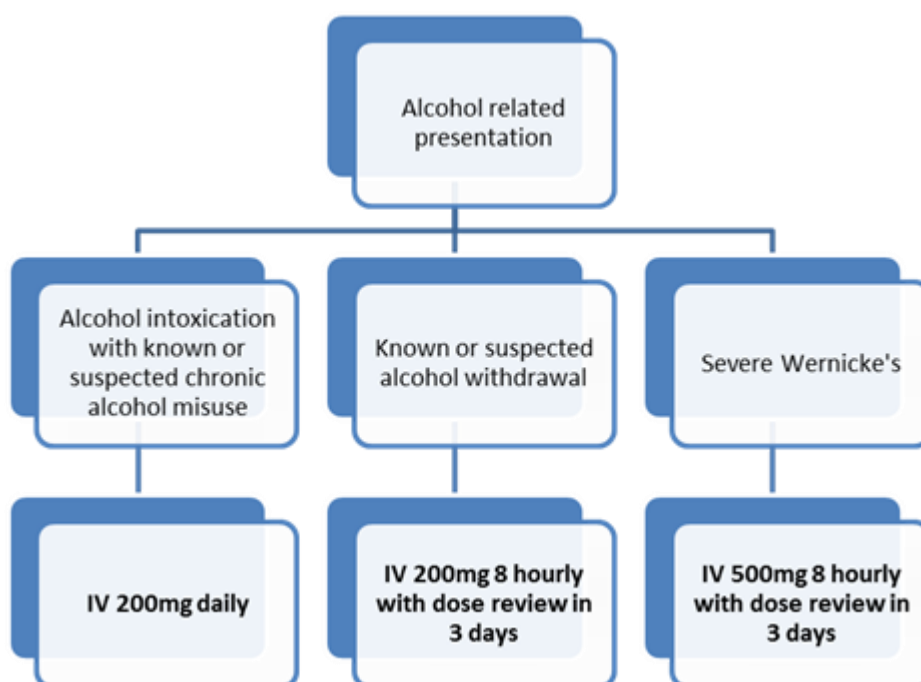
- Orally administered thiamine has poor bioavailability with a maximum of 4.5 mg available per 100mg ingested⁷. This is further reduced by up to 70% in the context of alcohol intoxication and malnutrition. Oral thiamine is therefore ineffective in treating thiamine deficiency and should not be used in the ED.
- Intravenous (IV) thiamine administration is the preferred route, although the intramuscular (IM) route is acceptable if there is no IV access. IM route is only suitable for doses up to 200mg.

IV glucose should not be given before thiamine administration as it may precipitate, WE in alcohol dependent patients^{6,9}.

Glucose supplements if required may be given after thiamine administration.

Magnesium is required for optimal utilisation of thiamine⁶. Levels should be checked and replenished if it is below normal range (0.70-1.10 mmol/L).

6.7.1. Thiamine dosing flowchart:



Doses of 200mg are diluted in 100mL of Sodium Chloride 0.9% and infused over 30 minutes.

Doses of 500mg are diluted in 100mL of Sodium Chloride 0.9% and infused over 60 minutes¹⁰.

- Additional thiamine treatment should be based on clinical response and should continue until there is no further clinical improvement.
- Following cessation of IV thiamine chronic drinkers with poor dietary intake and general poor nutritional status should be maintained on oral thiamine at 300mg per day for at least two weeks

- Oral thiamine supplementation (100 – 300mg PO daily) should be continued indefinitely in an alcohol dependent patient who continues to drink and should be considered on discharge¹¹.

6.8. Alcohol withdrawal chart

- The CIWA-Ar:
 - is validated for alcohol withdrawal only and should not be used to assess withdrawal from any other substance.
 - is not to be used as a diagnostic tool, as many other conditions produce similar signs and symptoms.
 - can be ceased when the patient has a total score of eight or less on four consecutive occasions once the patient is 48 hours post alcohol cessation.

7. Benzodiazepines

Benzodiazepines	Benzodiazepines are effective anxiolytic hypnotic medications. They are also effective amnestic agents and may have a disinhibitory effect especially at high doses.
Clinical Advisory Service	Is a specialist telephone support service provided by Next Step for health professionals providing clinical advice regarding the management of patients with AOD concerns.
Poly substance use	The use of two or three types of substances at the same time by a person.
Prescription shopping	When a patient visits multiple doctors for the purposes of procuring prescription medication without disclosing other consultations and/or prescriptions.
Prescription shopping programme (PSP)	A 24-hour prescription shopping information and alert service which assists prescribers to make informed prescribing decisions and better manage health outcomes for patients. A patient will meet the PSP criteria if within any three-month period they have been supplied with: • any PBS items prescribed by six or more different prescribers and /or • a total of 25 or more PBS target items and/or • a total of 50 or more PBS items (target and non-target).

7.1. Initial considerations

- Benzodiazepines are used in clinical practice for sedation and the management of anxiety, seizures, withdrawal states, insomnia and drug associated agitation.

- Benzodiazepines should only be used for short term treatment. Dependence develops relatively quickly, though this varies with each patient. Within 3-12 months of use 10-20% of patients may become dependent and this increases to 20-45% after more than a year of use⁹.
- Tolerance of different effects of benzodiazepines develops at different rates i.e. tolerance of sedative effects begins after two or three days and is significant by two to three weeks.
- Benzodiazepine use can result in a physical and psychological dependence and can be used in combination with other drugs.

7.2. Therapeutic dependence

- Many people take benzodiazepines prescribed from a single doctor for many years without dose escalation for management of anxiety or insomnia.
- Patients who use benzodiazepines may have difficulty ceasing use because they wish to avoid withdrawal and/or a return of the initial physical and/or psychological issues of concern.
- Risks associated with dependence are poorly defined however it is known that long term use is associated with an increased rate of cognitive decline and risk of falls. In this patient cohort the risk of continued use needs to be balanced against the distress and benefits of withdrawing.
- Patients in hospital for other reasons may undergo benzodiazepine withdrawal from even low doses of their regular long-term benzodiazepines. Dependant on use people may develop delirium and are also at risk of seizures during withdrawal.
- Withdrawal is best managed by very slow withdrawal supervised by the patient's General Practitioner (GP). Advice on strategies for optimising sleep and controlling anxiety should also be given.

7.3. Screening and assessment

- At assessment it is critical to identify the patient's use of benzodiazepines to determine how to manage their treatment / withdrawal.
- A history of benzodiazepine use is critical including collateral information from the GP and the PSP.
- The patient may be intoxicated on presentation which may impact their ability to provide and receive information.
- Assessment of intoxicated individuals should be limited to ensuring their safety. Caution should be taken when prescribing medications.
- When a patient meets the PSP criteria, verbal advice or a patient summary report on the PBS medicines dispensed and the number of prescribers involved during a defined period can be supplied.

- Further information on the PSP is available on the Australian Government Department of Human Services website.

7.4. Intoxication signs and symptoms

- Symptoms peak within 30 minutes to two hours of benzodiazepine use.
- Isolated intoxication with benzodiazepines presents with:
 - sedation where the patient responds to stimulation but rapidly relapses when not stimulated
 - slurred speech and drooling
 - blurred vision, headaches
 - loss of balance and coordination often associated with gait disturbance, muscle weakness or hypotonia
 - disinhibition and hypomania especially at high doses
 - depression, confusion, impaired cognition and memory (anterograde amnesia)
 - paradoxical euphoria, excitement, restlessness
 - nystagmus, vertigo
- Potentiation of other CNS depressants e.g. alcohol and opiates increase the likelihood of respiratory depression or death.

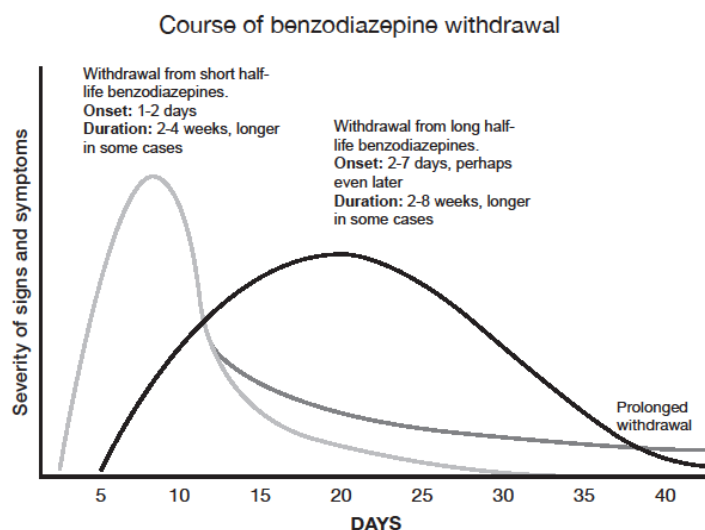
7.5. Withdrawal

- Benzodiazepine withdrawal is usually mild and symptoms are generally the same as those for which the benzodiazepines were initially prescribed. High dose use is more likely to present with severe symptoms.
- Where low-dose therapy is continued for six weeks or longer, tolerance and symptoms of withdrawal will affect 15-50% of patients.
- Withdrawal is more likely in:
 - usage exceeding three months
 - doses higher than therapeutically recommended
 - passive and dependent personality traits
 - sudden cessation of use.

7.6. Withdrawal onset

- Onset occurs two – five days after stopping, reaching a maximum on days seven – ten and usually abating by the end of the second or third week

Figure 2. Benzodiazepine withdrawal course



Adapted from Frank L., Peard J. New concepts in drug withdrawal: a resource handbook: ©1995 State of Victoria. Reproduced with permission.

- Withdrawal may occur earlier or later depending on the half-life of the benzodiazepine involved
- Withdrawal from long acting benzodiazepines may have less pronounced symptoms and the gradual decline in serum concentration may delay the onset of withdrawal by several days.

7.7. Withdrawal signs and symptoms

- Signs and symptoms of benzodiazepine withdrawal may include:
 - enlarged pupils, visual disturbance e.g. blurred vision, photophobia
 - fatigue, sweating, headaches
 - rebound insomnia and nightmares
 - loss of appetite, nausea, weight loss, diarrhoea
 - muscle twitching and pain, tremors
 - anxiety, panic attacks, depression, irritability, restlessness, agitation, delirium
 - generalised tonic-clonic seizures
- The major complication of benzodiazepine withdrawal is the development of delirium and seizures.

7.8. Withdrawal monitoring

- Benzodiazepine withdrawal scales are not used at FSFHG as there is little research to support their validity, reliability and predictive value.

7.9. Withdrawal management

- The aim of withdrawal management is to cease the benzodiazepine use safely without delirium or seizures.
- Wherever possible the community setting is the preferred option for managing withdrawal with:
 - an organised, structured and gradual regimen
 - GP involvement
 - linking to community alcohol and drug services to provide counselling and management as required.

7.10. Inpatient management

- Benzodiazepine use should not be abruptly discontinued even at low doses due to the potential risk of precipitating withdrawal and seizures.
- For patients with therapeutic dependence maintain benzodiazepines at preadmission levels.
- Patients on high doses or those who are poly-substance users should have a reduction plan initiated when medically stable.
- Refer to Section 12 for general nursing care requirements.

7.11. Pharmaceutical management of inpatient withdrawal

- Where inpatient withdrawal is required refer the patient to the appropriate AOD service or contact the AOD CAS.
- Withdrawal is best managed by stabilising the patient on one long acting benzodiazepine and then progressively reducing the dose.
- Patients taking high doses of benzodiazepines or poly-substance users should be stabilised on a long acting benzodiazepine (preferably diazepam) at a dose of approximately 40% of their regular intake before admission or 80mg/day whichever is the lower dose.
- Different benzodiazepines have different length of action and half-life. Refer to Table 4 Comparative oral doses of benzodiazepines.

Table 4. Comparative oral doses of benzodiazepines¹²

Drugs	Length of action	Half-life	Approximate equivalent dose to 5mg diazepam
Alprazolam	Short	6-12 hours	0.5mg
Bromazepam	Medium	12-24 hours	3mg
Clobazam	Long	> 24 hours	10mg
Clonazepam	Long	> 24 hours	0.25mg
Diazepam	Long	> 24 hours	5mg
Flunitrazepam	Long	> 24 hours	0.5mg
Lorazepam	Medium	12-24 hours	1mg
Nitrazepam	Long	> 24 hours	5mg
Oxazepam	Short	6-12 hours	15mg
Temazepam	Short	6-12 hours	10mg

- Contact the CAS when prescribing greater than 40mg of diazepam per day to discuss treatment options as the risk of overdose is higher.
- Reduction and withdrawal should be initiated once the patient is medically stable.
- Dose reductions should not exceed 10-20% per week and be titrated against symptoms. Rates of reduction need to be balanced with co-occurring issues such as anxiety which may be exacerbated during the withdrawal period.
- For patients stabilised on 40-80mg of diazepam daily withdrawal should be at a rate of:
 - at least 5mg per week until 40mg daily dose is achieved then
 - withdrawn at a rate of 2.5 mg per week thereafter e.g. reducing from 80mg diazepam daily at this rate would take nearly six months.
- The maximum rate of withdrawal in a patient stabilised on 40-80mg diazepam daily is:
 - to reduce the daily dose by 10mg at weekly intervals until the patient is on 40mg daily
 - then reduced by 5mg at weekly intervals, reducing from 80mg diazepam daily at this rate would take 12 weeks.
- Adjunct medications may be prescribed including:
 - Beta Blockers to control anxiety and autonomic hyperactivity

- Antidepressants for sedative effect (tricyclic antidepressants) or for the treatment of an underlying depressive disorder (serotonin reuptake inhibitors)
- Anticonvulsants may be prescribed for patients at increased risk of seizures i.e. chronic comorbid alcoholism or persistent high dose benzodiazepine users.
- The patient will be provided with clear information advising the aims of withdrawal and informing them they may feel uncomfortable and experience withdrawal symptoms.
- The reduction plan should be continued on discharge and the benzodiazepine dispensed daily with supervision at their local pharmacy.

8. Cannabis

Cannabis	The generic name given to the collection of materials derived from the plant cannabis sativa. Delta9-tetrahydrocannabinol (THC) is the psychoactive cannabinoid. It is classed as a hallucinogen as well as a depressant.
Cannabis hyperemesis syndrome (CHS)	A clinical syndrome of cyclical nausea and vomiting occurring after regular cannabis use. The mechanism of CHS is not well understood however, it is associated with regular cannabinoid use rather than being a withdrawal syndrome and is typically relieved with hot showers.
Psychotogenic	A substance that may cause a psychotic reaction.

8.1. Initial considerations

- Cannabis is considered psychotogenic and intoxication and withdrawal can be associated with precipitating or exacerbating underlying psychosis¹³.
- Underlying psychiatric illnesses or symptoms, which may have been masked during cannabis use, could emerge during the withdrawal period.
- Cannabis can be taken orally or can be smoked, often with tobacco, and the pharmacokinetics and pharmacodynamics of THC vary by route of exposure.
- Inhaled marijuana:
 - onset of psychoactive effects occurs rapidly with peak effects felt at 15-30 minutes and lasting up to four hours.
 - effects will mirror the plasma THC concentrations.
 - approximately 2-3 mg of inhaled THC is enough to produce drug effects in a naïve user.
- Ingested marijuana:
 - has a comparatively delayed onset of psychoactive effects that ranges from 30mins to three hours and clinical effects may last up to 12 hours.

- is less potent due to chemical degradation in gastric acid and poorer absorption.
- approximately 5-20mg of ingested THC is sufficient to produce drug effects in a naïve user.
- Several factors will influence how soon and for how long someone will feel intoxicated including patient weight, metabolism, gender and eating habits, tolerance levels, medications and other drugs
- Cannabis contains as many carcinogens as tobacco, deposits a third more tar and is thought to be a risk factor for oropharyngeal and lung cancer and coronary heart disease.

8.2. Intoxication signs and symptoms

- The signs and symptoms of cannabis intoxication include:
 - tachycardia, hypertension and tachypnoea
 - conjunctival injection (red eyes)
 - dry mouth
 - increased appetite
 - altered mental state, short term memory loss and time perception
 - disorientation, disinhibition, agitation, paranoia, visual or auditory hallucinations, anxiety and panic attacks
 - impaired coordination, cognition and judgement
 - seizures
 - hyperemesis.

8.3. Assessment and management of intoxication

- The patient may be intoxicated on presentation and this may affect their ability to provide and receive information. Where possible identify as accurately as possible:
 - dose, frequency, intensity, duration of use
 - method of consumption (smoking/ingesting)
 - poly drug use
 - co morbid physical or mental health issues
 - combination with nicotine – will also withdraw from nicotine
- Consider differential diagnosis or co-ingestion of other substances if severe symptoms persist.
- The management of cannabis intoxication involves supportive care and depends on clinical presentation.

- Perform and document vital signs on the appropriate Observation and Response Chart.
- Refer to Section 12 for general nursing care requirements.

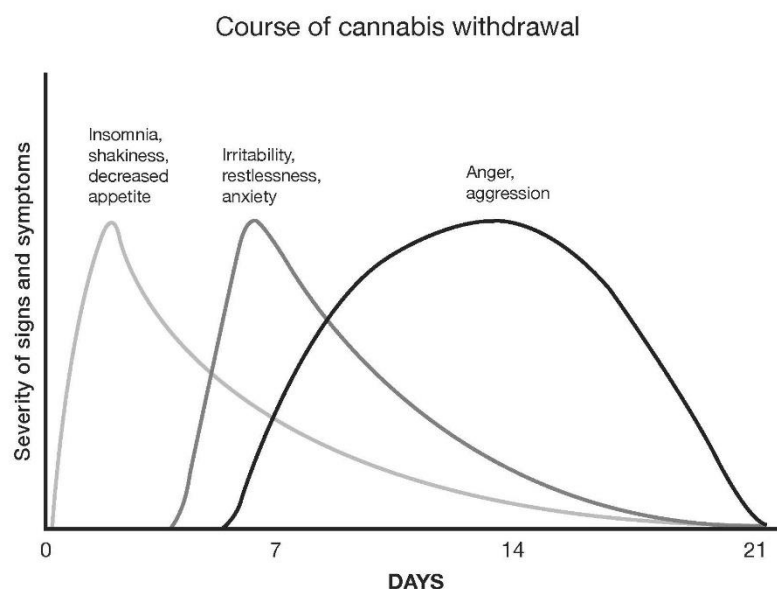
8.4. Cannabis withdrawal signs and symptoms

- Cannabis withdrawal is not life threatening and can be managed in the outpatient setting.
- Inpatient withdrawal may be considered if it is more conducive for the patient to withdraw on an inpatient ward due to psychosocial reasons.
- Symptoms are variable and may not relate to the quantity smoked or duration of use; consideration should be given to the following variables which may impact the severity of withdrawal including:
 - psychiatric comorbidity
 - quantity and potency of preparation consumed
 - premorbid tendencies toward aggression and violence
 - duration of current use
 - rate of reduction in use
 - comorbid substance use
 - physical health and current medications.
- Studies suggest 50-75% of dependent cannabis users will experience four or more withdrawal symptoms.

Table 5. Symptoms of cannabis withdrawal

Common symptoms	Less common/ equivocal symptoms
Anger or aggression	Chills
Decreased appetite or weight loss	Depressed mood
Irritability	Gastrointestinal tract symptoms e.g. abdominal cramps
Nervousness/ anxiety, agitation, restlessness	Shakiness, tremor
Nausea and vomiting	Diaphoresis (sweating), night sweats
Disturbed sleep and vivid dreams	Hiccups
Cravings	Stuffy nose
	Fatigue
	Yawning

Figure 3. Course of cannabis withdrawal



- Most withdrawal symptoms:
 - emerge one to two days after last use
 - peak within the first seven days
 - continue for approximately seven to 14 days¹⁴.
- Different symptoms may peak at different times throughout the withdrawal e.g. anxiety, anger and aggression.
- Cannabis withdrawal scales are not used at FSFHG as there is little research to support their validity, reliability and predictive value
- Cannabis withdrawal syndrome may last for several weeks and patients should be referred to a drug and alcohol service for ongoing support.

8.5. Cannabis withdrawal management in the inpatient setting

- For cannabis withdrawal in the inpatient setting refer the patient to the appropriate AOD service or contact the AOD CAS as required.
- Patients should be involved in decisions and treatment planning regarding interventions around their cannabis use and be provided with accurate information and harm minimisation advice.
- Supportive therapies for psychological withdrawal may include counselling, motivational interviewing, harm reduction, relapse prevention, sensory modulation, cognitive behavioural therapies and other psychosocial interventions and self – help groups.
- Two cannabis specific reduction strategies are:

- sudden cessation, with or without symptomatic relief
- gradual dose reduction using diaries in combination with evidence based psychological intervention.
- Many cannabis users will be able to stop cannabis use abruptly without experiencing any significant physical withdrawal symptoms.
- Provide symptomatic relief as required, most clinical practice involves the use of benzodiazepines.

Table 6. Pharmaceutical treatments for symptomatic relief⁶

Cannabis Withdrawal Symptoms	Suggested pharmaceutical treatments for symptomatic relief
Nausea & vomiting	• Ondansetron PO/sublingual/IV 4-8mg TDS PRN. • Metoclopramide PO/IM/IV 5-10mg TDS/PRN Maximum five days of treatment • Prochlorperazine PO 5mg, every 4-6 hours as required
Muscle aches & pains and headaches	• Paracetamol 1g QID PRN PO
Agitation and/or anxiety, restlessness, irritability	• Diazepam PO 5mg up to QID PRN (adjust dose to clinical need and risks) • Acute short-term use only, up to seven days • MH patients refer to the FSFHG Medication Management of Agitation and Arousal in Mental Health Inpatients Guideline
Insomnia	• Temazepam 10mg nocte PRN PO • Maximum of seven days treatment
Abdominal cramps	• Hyoscine butylbromide 10-20mg TDS PRN PO/IV
Symptoms of psychosis e.g. thought disorders, paranoia, perceptual disturbances	• Consider the use of a benzodiazepine • Consider oral antipsychotic medications to manage psychotic features as clinically required.

- Seek advice from AOD service or Specialist Medical Practitioner if further treatment options are required.

8.6. Pregnancy and breastfeeding

- Studies on the effects of maternal cannabis use have been inconsistent. It has been suggested that cannabis use during pregnancy is not an independent risk factor for adverse neonatal outcomes¹⁵. However, cannabis use may be associated with the use of other substances which can be harmful to the neonate e.g. tobacco.
- Pregnant women are strongly recommended to cease cannabis and tobacco use.
- Cannabis use is not recommended in breastfeeding
- Consider referring patients to WANDAS for additional specialist advice and ongoing support.

8.7. Cannabis Hyperemesis Syndrome (CHS)

- Signs and symptoms of CHS include:
 - frequent requests for hot showers
 - severe nausea and vomiting
 - abdominal pain with negative medical workup
 - dehydration and pre-renal acute kidney failure
- Management includes:
 - cessation of cannabis is the only definitive treatment for CHS
 - supportive treatment including fluid resuscitation for dehydration
 - antiemetic e.g. Metoclopramide or Ondansetron if required.
 - Capsaicin cream 0.025-0.075% applied topically for symptoms of hyperemesis.

9. Opioids

Opioids	The generic term applied to alkaloids from the opium poppy, their synthetic analogues, and similar compounds synthesised within the body. Includes heroin, fentanyl, oxycodone, hydrocodone, codeine, morphine, methadone, buprenorphine and others.
Clinical Opiate Withdrawal Scale (COWS)	An 11-item objective scale designed to measure severity of opioid withdrawal symptoms.

9.1. Use of opioids

- Prolonged use of opioids can lead to the development of tolerance and dependence¹⁶.

- Cessation after prolonged use is associated with a range of withdrawal symptoms. Withdrawal syndrome is uncomfortable but not usually life threatening¹⁶.
- Tolerance to opioids decreases rapidly following withdrawal which increases the risk of opioid overdose if the client resumes prior levels of opioid use¹⁶.
- Harm minimisation advice and ongoing support are crucial to reducing the likelihood of early relapse with potentially fatal consequences.
- Patients not ready for complete withdrawal should consider an opioid substitution treatment program.

9.2. Intoxication signs and symptoms

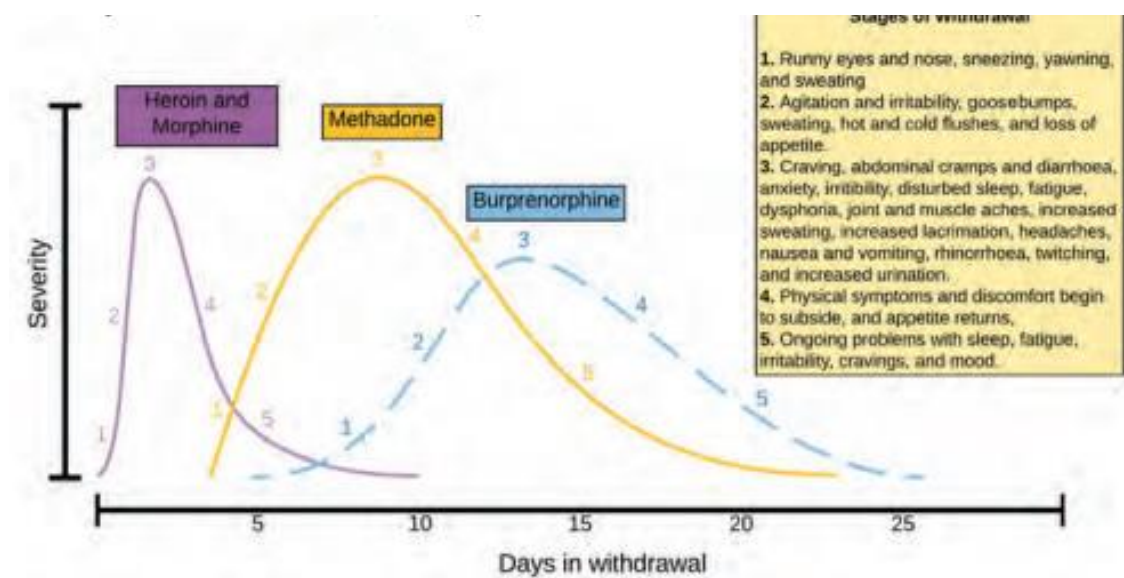
- Opioid intoxication causes central nervous system (CNS) depression and death may result secondary to respiratory failure.
- Classic opioid toxidrome includes:
 - CNS depression
 - respiratory depression
 - miosis (pupil constriction).
- Some opioids may have unexpected toxicity:
 - Methadone use is associated with QT prolongation¹⁷
 - Pethidine¹⁸/Fentanyl¹⁹ use may cause serotonin syndrome.

9.3. Overdose management

- Patient care includes:
 - airway, breathing, circulation management refer to FSH & FH Code Blue medical emergency procedures
 - Naloxone is the specific antidote. Titrated doses of intravenous naloxone in line with FSFHG Naloxone administration guideline
 - consult toxicology services
 - consider Naloxone prefilled take-home syringes (Prenoxad) and nasal delivery device (Nyxoid) on discharge.

9.4. Opioid withdrawal

- Opioid withdrawal is the physiological response that results from the abrupt cessation or rapid reduction in opioid dose in an opioid dependent person or when the person is administered an opioid antagonist or partial agonist.
- The aim of withdrawal treatment is to support the patient and provide symptomatic relief from discomfort.
- The onset and duration of withdrawal depends on the half-life of the drug being taken.
- The severity and time course of withdrawal is dependent on the opioid used.

Figure 4. Time course and stages of opioid withdrawal⁵

Source: NSW Department of Health 2008

Table 7. Opioid withdrawal symptoms¹⁶

Common symptoms and time frame of opioid withdrawal	
Stage of Withdrawal	Common Symptoms
Early	• Runny eyes and nose, sneezing, yawning • Diaphoresis (sweating), hot and cold flushes • Loss of appetite • Goosebumps
Peak	• Strong cravings • Stomach cramps and diarrhoea • Nausea and vomiting • Body aches • Restlessness, agitation, irritability, anxiety and disturbed sleep • Lacrimation, rhinorrhoea, twitching • Lethargy and poor concentration • Hot and cold flushes with increased sweating • Increased urination
Late	• Physical symptoms begin to subside • Psychological symptoms such as lethargy, irritability, cravings and insomnia may persist but in lower severity

9.5. Opioid withdrawal management

- A comprehensive assessment and history are crucial in helping formulate a treatment plan including withdrawal management.
- Patients will be regularly monitored using the COWS.
- Assess for physical health harms which may occur with opioid use e.g. history of intravenous drug use has higher risk of BBVs.
- A liver function test may be beneficial for patients who have been on long term codeine and paracetamol combinations.
- Refer to Section 12 for general nursing care requirements.

9.6. Clonidine use for management of withdrawal

- Clonidine is accepted for the management of opioid withdrawal symptoms in Australia. It can be used to reduce withdrawal symptoms such as nausea, vomiting, agitation and sweating¹⁶.
- Clonidine has little effect on reducing cravings; however, it does not produce opioid like tolerance or dependence.
- Caution should be used when administering Clonidine with other depressant medications e.g. Benzodiazepines.
- Patients taking Clonidine require monitoring for adverse effects including:
 - hypotension
 - bradycardia
 - sedation
- Prior to administering first dose of clonidine:
 - record baseline blood pressure and heart rate
- Clonidine must **not** be administered if:
 - the patient is pregnant
 - systolic blood pressure < 90mmHg or
 - diastolic blood pressure < 50mmHg or
 - heart rate < 50 beats per minute or
 - clinical evidence of impaired circulation
- Initial clonidine test dose:
 - give 75 – 100micrograms orally and monitor blood pressure at 30 minutes post administration and document on the appropriate Observation and Response Chart
 - lying and standing blood pressure is measured pre and post
 - further Clonidine is not recommended if systolic blood pressure is <90mmHg, diastolic blood pressure < 50mmHg, heart rate < 50 beats per minute

- if no adverse side effects e.g. hypotension, dizziness a further 75 – 100micrograms oral dose of clonidine may be administered and continue treatment as per the table of symptomatic treatments below.
- Maximum dose of 800micrograms daily.

Table 8. Symptomatic treatment management for mild opioid withdrawal

Symptoms	Suggested treatments
Nausea and vomiting	• Ondansetron PO/sublingual/IV 4-8mg TDS PRN. • Metoclopramide PO/IM/IV 5-10mg TDS/PRN Maximum 5 days of treatment • Prochlorperazine PO 5mg, every 4-6 hours PRN
Muscle aches and Pain	• Paracetamol PO 500-1000mg every 4-6 hours PRN. Maximum 4000mg Paracetamol in 24 hours • Ibuprofen PO 400mg every 6-8 hours PRN
Agitation and/or anxiety	• Clonidine PO 75 -100microgram every 6 hours • Diazepam PO 5mg QID PRN • MH patients refer to the FSFHG Medication Management of Agitation and Arousal in Mental Health Inpatients guideline
Insomnia	• Temazepam PO 10-20mg PRN at night
Abdominal cramps	• Hyoscine butylbromide PO/IV 10-20mg every six hours PRN • Paracetamol may also provide some relief
Diarrhoea	• Loperamide PO 2mg PRN after loose bowel motion. Maximum 16mg in 24 hours

9.7. Pregnancy

- Management of opioid dependence/withdrawal in pregnancy is a highly specialised area. Refer pregnant patients to WANDAS for additional specialist advice and ongoing support.
- Opioid withdrawal during pregnancy is a high-risk option as relapse will potentially expose both mother and child to illicit substance use.

- Withdrawal prior to 14 weeks gestation or after 32 weeks gestation is associated with a higher risk of:
 - miscarriage
 - foetal distress, death
 - neonatal withdrawal
- Opioid substitution treatment (OST) is the preferred management for opioid dependent pregnant women¹⁶.
- Compared to continued illicit drug use or withdrawal, OST is associated with improved foetal development, infant birth weight, and reduced risk of perinatal and infant mortality¹⁶.
- Patients who decline an OST program should be encouraged to consider withdrawal in the 2nd trimester.
- Risk vs benefit scenarios should be clearly explained to the patient.
- The use of Buprenorphine only (Subutex®) and Buprenorphine/naloxone (Suboxone®) is considered safe in all trimesters. Both are considered safe in breastfeeding. Refer to the King Edward Memorial Hospital [Community Programme for Opioid Pharmacotherapy \(CPOP\) Management in Hospital Clinical Practice Guidelines](#) for more information.

9.8. Buprenorphine/Naloxone (Suboxone®) withdrawal protocol

- All patients undergoing this opioid detoxification within the hospital should be referred to and/or reviewed by the AOD service to ensure appropriate assessment of suitability, to provide wrap around support and referral to long term opioid substitution therapy on discharge.
- For patients with a heavy opioid dependence who are likely to go through moderate to severe withdrawal whilst in a public hospital; specialist medical practitioners specified under Section 2.5.8 of the [Schedule 8 Medicines Prescribing Code](#) can prescribe detoxification therapy.
- The patient must be made aware that treatment with Suboxone® for opioid detoxification will result in them being included on the Drugs of Addiction Record as a drug dependent person.
- It is the responsibility of the treating specialist prescriber to discuss the implications of detoxification treatment with the patient and also provide the patient with a copy of the Department of Health "[Patient Information: Reporting Drug Dependent Persons](#)" information. Reports of drug dependence must be made on the '[Reporting a Drug Dependent Person Form](#)'
- Prior to commencing therapy, the specialist medical practitioner must complete and submit the 'Notification of the use of Suboxone®' Form for the treatment of opioid withdrawal in a hospital setting. This form is available by contacting CAS and requesting a copy of the notification form.

- Send the completed 'Notification of the use of Suboxone®' Form for the treatment of opioid withdrawal in a hospital setting to:
 - CPPAdmin@mhc.wa.gov.au or fax: 9471 0444.
 - After hours or on weekends – to Next Step On-Call Doctor
- If accepted by the Department, the applicant will become an approved detoxification prescriber for inpatients at the public hospital specified in the form.
- A notification form must be submitted for each patient.
- The detoxification period must be short term and may not exceed 5 days for any single period using the approved buprenorphine (Suboxone®) detoxification protocol (below).
- Changes to the approved protocol will require prior written authorisation from the Health Department and DACAS must be contacted for approval.

Table 9. Approved Buprenorphine (Suboxone) Protocol:

Day	Suboxone® Film Regimen	Recommended Total Daily dose
Day 1	4mg at onset of withdrawal (*COWS score ≥ 12) Potential additional 2 to 4mg evening dose if necessary	4mg (maximum 8mg)
Day 2	4mg mane Potential additional 2 to 4mg evening dose if necessary	4mg (maximum 8mg)
Day 3	4mg mane Potential additional 2mg evening dose if necessary	4mg (maximum 6mg)
Day 4	2mg mane Potential additional 2mg evening dose if necessary	2mg (maximum 4mg)
Day 5	2mg if necessary	0mg (maximum 2mg)
Patient's COWS Score (see COWS form) Date: _____ Time: _____ *COWS score should be assessed as ≥ 12 prior to Suboxone® administration		

- To reduce the risk of diversion the patient must place the prescribed dose of Suboxone® film underneath the tongue, taking care not to overlap multiple films and the nurse must supervise the patient for one minute after the dose is administered.
- Patients must not be supplied with Suboxone® to self-administer later, or at a different facility, or outside the observation of an approved health practitioner.
- During detoxification, a patient may not be prescribed, supplied or administered any other S8 medicine without the permission of the Chief Executive Officer. All reasonable steps must be taken to prevent patient access to other S8 medicines during this period. The authorised specialist must inform other S8 prescribers where known, when detoxification commences.

10. Psychostimulants (amphetamines, cocaine and ecstasy)

Psychostimulants /Amphetamine type substances (ATS)	A group of drugs which can stimulate the sympathetic and the central nervous system by increasing the synaptic concentration of excitatory neurotransmitters dopamine, norepinephrine and serotonin¹⁶. ATS can increase the release of these neurotransmitters, inhibit their reuptake or do a combination of both to produce a range of stimulatory effects. Refers to a group of drugs which includes: • Amphetamines • Methylenedioxymethamphetamine (MDMA) • Cocaine hydrochloride A range of psychostimulants are also available for the treatment of various conditions. These may be used illicitly including: • Methylphenidate (Ritalin ® and Concerta ®) • Phentermine (Duromine ® and Metermine ®) • Pseudoephedrine (Dimetapp ®, Sudafed ®)²
Amphetamines	Includes the four types of amphetamines: • Amphetamine • Dexamfetamine • Lisdexamfetamine • Methamphetamine
Sympathomimetic toxidrome	A group of symptoms associated with poisoning or exposure to a substance or class of substances including delusions, paranoia, tachycardia, hypertension, hypotension, hyperthermia, sweating and seizures.

10.1. Use of psychostimulants

- In general stimulants increase alertness, the sense of wellbeing and the pleasurable reinforcement experienced with many activities.

10.2. Signs and symptoms of intoxication

Table 10. Signs and symptoms of psychostimulant intoxication¹⁶

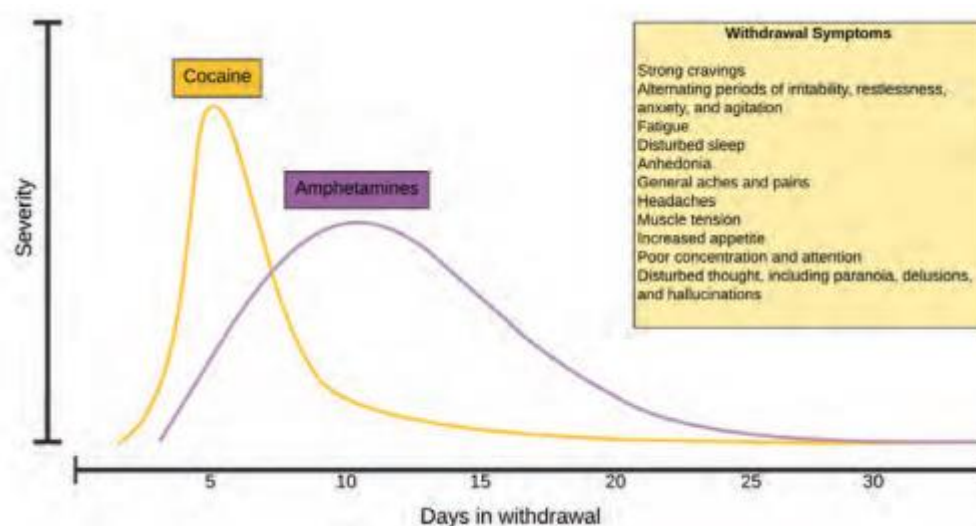
System	Symptoms
CNS, neurological and behavioural	• overstimulation, restlessness, increased energy, reduced fatigue and insomnia

	• heightened alertness, headache, mydriasis, dizziness and tremors • overconfidence, violent, unpredictable or irrational behaviour • euphoria • pressured speech • bruxism (teeth grinding) • seizures and coma • anorexia (appetite suppression) • confusion and psychosis (hallucinations, delusions and paranoia)
Cardiovascular	• hypertension • palpitations • tachycardia, arrhythmia • angina, acute coronary syndrome, aortic dissection
Gastrointestinal	• nausea and vomiting • abdominal cramps, diarrhoea or constipation
Other	• tachypnoea (increased respiration), respiratory failure • pyrexia (increased temperature) • flushing or pallor • diaphoresis (sweating)

10.3. Withdrawal

- A thorough assessment will include:
 - type of stimulant and its potency
 - dose, frequency, intensity, duration of use
 - mode of administration
 - poly substance use
 - comorbid physical or mental health issues
- Withdrawal can commence in the ED with identification and risk assessment and continued over the length of the admission.

- Withdrawal is not life threatening and it is safe for a patient to withdraw in the community setting. A biopsychosocial assessment should be completed when considering the best setting for withdrawal.
- Withdrawal management aims to manage symptoms and engage the patient in psychoeducation and harm reduction.
- Patients who present solely to reduce or cease their psychostimulant use will be referred to specialist outpatient AOD services to support reduction, cessation and harm reduction where inpatient admission is not indicated.

Figure 5. Phases of stimulant withdrawal⁵

Source: NSW Department of Health 2008

Table 11. Psychostimulant withdrawal¹⁶

Phase	Time since last stimulant use	Common signs and symptoms
Crash*	• Starts 12 to 24hrs after last use and subsides by day 2 to 4. • These symptoms are more pronounced following high dose use.	• exhaustion and fatigue • overwhelming desire to sleep or experiencing sleeping disturbances • fluctuating emotional state – dysphoric, and can be associated with anxiety or agitation • low cravings • generalised aches and pains • hunger

Withdrawal*	• Generally, commences 2 to 4 days after last use • peaks in severity over 7 to 10 days • can persist for 6 to 10 weeks	• strong cravings and urge to use • sleeping difficulties and disrupted sleeping patterns • poor concentration and attention • fluctuations in mood and energy levels; alternating between irritability, restlessness, anxiety and agitation • headaches and generalised aches and pains • increased appetite • thought disturbance (e.g. predominantly paranoia) and perceptual disturbance (e.g. hallucinations) which may have been masked in the crash phase • perceptual disturbance (e.g. hallucinations) which may have been masked during the crash phase
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10.4. Withdrawal management

- Management of acute presentations includes:
 - identifying and treating symptoms/complications of psychostimulant misuse
 - providing a safe patient environment
 - patient observation and behavioural management
 - medical and psychiatric assessment as required
 - engagement with AOD services
 - for Mental health patients refer to the Medication Management of Agitation and Arousal in Mental Health Inpatients guideline
- Consider Sympathomimetic toxidrome management.
- Refer to Section 12 for general nursing care requirements.

10.5. Psychostimulant use in pregnancy

- Psychostimulant use during pregnancy is associated with harms to the mother and neonate including an increased risk of impaired neonatal growth, preterm labour and delivery. Psychostimulant use can affect the overall development of the baby¹⁶.
- Hypertension resulting from psychostimulant use can increase the risk of antepartum haemorrhage, cardiovascular accident or acute coronary syndrome¹⁶.

- Withdrawal from ATS during pregnancy is not dangerous – any medication prescribed should be considered in terms of benefit to mother and minimising the risk to the pregnancy¹⁶.
- Referral to the WANDAS for specialist advice and ongoing support should be discussed and encouraged with the patient to ensure best care for both mother and baby.

11. Volatile substances

Volatile substances	Includes various hydrocarbons, ethers, ketones and alkyl halides which are common ingredients in household products e.g. petrol, paint thinner, glue, lighter fluid.
Volatile substance use (VSU)	Refers to the practice of deliberately inhaling substances that are vaporous at ambient temperatures for becoming intoxicated.
Street /slang terms	Methods of use are commonly referred to as 'huffing', 'sniffing', 'chroming', 'bagging', and 'nanging'

11.1. Intoxication risks

- Acute risks include:
 - sudden death due to hypoxia
 - cardiac sensitisation and dysrhythmia
 - micro aspiration and pneumonitis
- Long term risks include:
 - neuropsychiatric impairment
 - renal tubular acidosis from toluene
 - subacute combined degeneration of spinal cord from vitamin B12 deficiency due to nitrous abuse.

11.2. Intoxication Signs and Symptoms

- Intoxication usually occurs within one to five minutes after initial inhalation and can last for several minutes after the person stops inhaling.
- When a person continues to inhale the effects can potentially last for hours even after the person ceases to inhale.
- Signs and symptoms may include:
 - euphoria/excitability
 - reduced inhibitions/impaired judgement, attention and memory
 - dizziness, headache

- blurred vision, slurred speech
- central nervous system depression slowed heartrate & breathing, drowsiness and sedation
- disorientation/confusion
- hallucinations, delusions or delirium.

11.3. VSU withdrawal

- People who use inhaled volatile substances may experience a withdrawal syndrome.
- Generally, symptoms are not severe and last only two to five days in most people, up to one week in heavy chronic users.
- Symptoms of withdrawal include:
 - anxiety/irritability
 - restlessness/agitation
 - nausea and vomiting
 - diarrhoea, abdominal pain
 - tremor, perspiration.

11.4. Clinical management

- Most cases of acute VSU intoxication can be managed by removing the substance and encouraging rest.
- Management requires treatment of symptoms as necessary.
- Sedatives should only be used if required for acute behavioural disturbances.

12. General nursing care

The following general nursing care applies to all substances.

12.1. Mobility

- A falls risk assessment will be completed in line with the FSFHG Falls prevention and management policy.
- Consider initially restricting the patient to bed rest.
- Ambulate with supervision when assessed as safe to do so.
- Supervise as indicated with activities of daily living.
- Assist with showering/hygiene needs if required.
- Consider referral to physiotherapy for assessment.

12.2. Diet and hydration

- Administer intravenous fluids as prescribed

- Encourage oral fluids to correct dehydration when alert and orientated.
- Consider commencement of fluid balance chart to monitor intake and output.
- Discourage caffeine where possible.
- Encourage small, regular meals if clinically appropriate.

12.3. Skin integrity

- Care is provided in line with the FSFHG Pressure injury prevention and management policy.

12.4. Involving the patient, family and carers

- Provide patients and families with a treatment setting and experience that is as welcoming, accepting, non-judgemental and responsive¹ as possible.
- Involve patients in decision making and treatment planning.
- Engage family members and people who are significant to the patient in the treatment process (with the patient's approval)¹.
- Provide patient/family with relevant information and resources.

12.5. Follow up / Discharge planning

- Involve patient/family in discharge planning informing them of their options in the community where these supports are available.
- Facilitate engagement with community and/or specialist support service providers who will provide ongoing care where possible
- Include information in the discharge plan on how the patient / family can seek additional support including return to the current service.
- Provide information on strategies to manage and reduce risks or harms associated with continued substance use.
- The patients discharge summary must be completed within 24 hours of discharge and forwarded to the patients GP.

13. Compliance/performance monitoring

Medical and Nursing Directors will be responsible for identifying and implementing compliance monitoring strategies appropriate to their area. These compliance monitoring strategies may include evaluation, clinical incident reporting, auditing and/or feedback processes.

Identified compliance measures will be reported to the Service Safety Quality Risk Committee.

14. Related policy documents

WA Health:

- [Alcohol and other drug withdrawal management policy \(MP0062/17\)](#)
- [WA Health Consent to Treatment Policy](#)
- [Medicines Handling policy MP 0139/20](#)
- [The Schedule 8 Medicines prescribing code](#)
- [Recognising and responding to acute deterioration](#)
- [Alcohol and Other Drugs Partnerships and Pathways Version 1.0](#)
- [Alcohol and Other Drugs Early Intervention Practice and Pathways Version 1.0](#)
- [Walk with me project Recommendation Report](#)

SMHS:

- [Alcohol and Prohibited Substances](#) (SMHS: 112)
- [Testing mental health inpatients for alcohol and/or illicit substance use](#) (SMHS: 151)

FSFHG:

- [Prevention and management of unpredictable behaviours](#) (FSFH-MENH-POL-0026)
- [Naloxone prefilled take-home syringes \(Prenoxad\) and nasal delivery device \(Nyxoid\)](#) (FSFH-HW-GUI-0029)
- [Pressure Injury Prevention and Management](#) (FSH-HW-POL-0025)
- [Medication Administration](#) (FSFH-HW-POL-0022)
- [Observation and Response Chart](#) (FSFH-HW-POL-0032)
- [Management of Agitation in Older Adults with Dementia or Delirium](#) (FSFH-HW-GUI-0006)
- [Cognitive screening of the older person](#) (FSFH-HW-POL-0028)
- [Falls Prevention and Management](#) (FSFH-HW-POL-0027)
- [Thiamine - Intravenous Drug Administration Guideline](#)
- [Emergency Department adult bolus guidelines: Resuscitation](#) (FSFH-ED-GUI-003)
- [Next Step Inpatient Withdrawal Unit \(IPWU\) Emergency Referral Pathway](#) (FSH-ED-PRO-0014)
- [Decompensated cirrhosis: Early Management](#) (FSFH-GAST-GUI-0001)
- [Medication Management of Agitation and Arousal in Mental Health Inpatients guideline](#) (FSFH-MENH-GUI-0004)

- FSH Code Blue - Medical Emergency
- FH Code Blue - Medical Emergency

15. Related Documents

- [Mental Health Act 2014](#)
- [Medicines and Poisons Act 2014](#)
- [Medicine and Poisons Regulations 2016](#)
- [Health \(Notification of Stimulant Induced Psychosis\) Regulations 2010](#)
- [Working together - Mental Health and Alcohol and Other Drug Engagement Framework 2018-2025.](#)

16. Related standards

NSQHS standards:

- Partnering with Consumers
- Medication Safety
- Comprehensive Care
- Communicating for safety

NSMHS standards:

- Promotion and Prevention
- Integration
- Delivery of care

Chief Psychiatrists Standards for Clinical Care

- Assessment
- Care Planning

17. References

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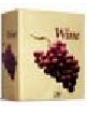
18. Authorisation

EXECUTIVE SPONSOR: Nurse Director, Service 5					
Version	Date Issued	Compiled/Revised By	Committee/Consumer Group Consulted	Endorsed By	Revision due
1	11/2019	CNC, AOD	DTC and NMPGC	FSFHG Policy Committee	11/2022
2	02/2020	Senior Pharmacist	DTC	FSFHG Policy Committee	11/2022
3	04/2021	CNC, AODS	DTC	FSFHG Policy Committee	11/2022
4	08/2021	CNC, AODS	DTC	FSFHG Policy Committee	11/2022
5	02/2023	CNC, AODS	DTC, NMPGC	FSFHG Policy Committee	02/2026

19. Appendices

19.1 Appendix 1: Standard drinks guide

NUMBER OF STANDARD DRINKS – BEER								
								
1.1	0.8	0.6	1.6	1.2	0.9	1.4	1	0.8
285ml	285ml	285ml	425ml	425ml	425ml	375ml	375ml	375ml
Full Strength	Mid Strength	Low Strength	Full Strength	Mid Strength	Low Strength	Full Strength	Mid Strength	Low Strength
4.8% Alc. Vol	3.5% Alc. Vol	2.7% Alc. Vol	4.8% Alc. Vol	3.5% Alc. Vol	2.7% Alc. Vol	4.8% Alc. Vol	3.5% Alc. Vol	2.7% Alc. Vol
								
1.4	1	0.8	34		24		19	
375ml	375ml	375ml	24x375ml		24x375ml		24x375ml	
Full Strength	Mid Strength	Low Strength	Full Strength		Mid Strength		Low Strength	
4.8% Alc. Vol	3.5% Alc. Vol	2.7% Alc. Vol	4.8% Alc. Vol		3.5% Alc. Vol		2.7% Alc. Vol	

NUMBER OF STANDARD DRINKS – WINE						
						
1.6	1	0.9	1.4	1	1.4	7.5
150ml	100ml	60ml	150ml	100ml	150ml	750ml
Average Restaurant Serving of Red Wine	Standard Serve of Red Wine	Standard Serve of Port	Average Restaurant Serving of White Wine	Standard Serve of White Wine	Average Restaurant Serve of Champagne	Bottle of Champagne
13.5% Alc. Vol	13.5% Alc. Vol	18% Alc. Vol	11.5% Alc. Vol	11.5% Alc. Vol	12% Alc. Vol	12.5% Alc. Vol
						
8	43	21	7.5	39	19.5	20
750ml	4Litres	2 Litres	750ml	4 Litres	2Litres	2Litres
Bottle of Red Wine	Cask Red Wine	Cask Red Wine	Bottle of White Wine	Cask White Wine	Cask White Wine	Cask of Port
13.5% Alc. Vol	13.5% Alc. Vol	13.5% Alc. Vol	12.5% Alc. Vol	12.5% Alc. Vol	12.5% Alc. Vol	17.5% Alc. Vol

NUMBER OF STANDARD DRINKS– SPIRITS							
							
1	22	1.1	1.2	2.6	1.5	1.8	3.6
30ml	700ml	275ml	330ml	660ml	275ml	330ml	660ml
High Strength Spirit Nip	High Strength Bottle of Spirits	Full Strength RTD*	Full Strength RTD*	Full Strength RTD*	High Strength RTD*	High Strength RTD*	High Strength RTD*
40% Alc. Vol	40% Alc. Vol	5% Alc. Vol	5% Alc. Vol	5% Alc. Vol	7% Alc. Vol	7% Alc. Vol	7% Alc. Vol
							
1	1.2	1.5	1.7	1.4 – 1.9	1.6	2.1	2.4
250ml	300ml	375ml	440ml	250ml	300ml	375ml	440ml
Full Strength Pre-mix Spirits	Full Strength Pre-mix Spirits	Full Strength Pre-mix Spirits	Full Strength Pre-mix Spirits	High Strength Pre-mix Spirits	High Strength Pre-mix Spirits	High Strength Pre-mix Spirits	High Strength Pre-mix Spirits
5% Alc. Vol	5% Alc. Vol	5% Alc. Vol	5% Alc. Vol	7% – 10% Alc. Vol	7% Alc. Vol	7% Alc. Vol	7% Alc. Vol

19.2 Appendix 2: Drugs with the potential to interact with alcohol

Medication	Type of interaction
Sedative–hypnotics: • benzodiazepines • barbiturates	• Acute alcohol consumption potentiates the central nervous system depressant effects of benzodiazepines and barbiturates. • Chronic alcohol consumption decreases availability of barbiturates through hepatic enzyme induction, decreasing their effect. • Increased risk of cognitive impairment, respiratory depression and overdose.
Anaesthetic agents	• Chronic alcohol consumption: • Increases the dose of propofol required to induce anaesthesia • Increases the risk of liver damage by anaesthetic gases enflurane and halothane.
Opioid analgesics	• Alcohol increases sedative effect. • Increased risk of cognitive impairment, respiratory depression and opioid overdose.
Tricyclic antidepressants	• Acute alcohol consumption increases risk of sedation and orthostatic hypotension (sudden drop in blood pressure upon standing up).
Antihistamines	• Alcohol potentiates the central nervous system depressant effect of sedating antihistamines, especially in elderly people.
Antipsychotic medication: • Phenothiazines • Olanzapine	• Acute alcohol consumption increases sedative effects, impairs coordination and may result in liver impairment. • Alcohol increases sedation and risk of hypotension.
Oral hypoglycaemic agents: • Sulfonylurea compounds	• Diabetics on sulfonylureas should be advised not to drink. • Acute alcohol ingestion prolongs availability of hypoglycaemic agents leading to hypoglycaemia. • Hypoglycaemia may also occur if there is malnutrition or depletion of glycogen stores. • Chronic alcohol administration decreases the availability of hypoglycaemic agents with risk of hyperglycaemia.

Anticonvulsants: Phenytoin	- Acute alcohol consumption increases availability of Phenytoin increasing risk of side effects. - Chronic alcohol consumption decreases anticonvulsant effect of Phenytoin.
Histamine H2 receptor antagonists: Cimetidine	Inhibit gastric alcohol dehydrogenase and increase the rate of gastric emptying. This may increase blood alcohol concentration.
Oral anticoagulants: Warfarin	- Acute alcohol consumption increases Warfarin's availability, increasing risk of haemorrhage. - Chronic alcohol consumption reduces availability of Warfarin, decreasing its anticoagulant effect.
Non-narcotic analgesics: - Aspirin, NSAIDs - Paracetamol	- Alcohol increases the risk of gastrointestinal bleeding. - Chronic alcohol consumption increases risk of liver damage with paracetamol overdose.
Antibiotics: Metronidazole	- Metronidazole appears to inhibit alcohol dehydrogenase and other alcohol oxidising enzymes. - Patients should be advised not to take alcohol during therapy or for at least 48 hours afterwards because of the possibility of a disulfiram-like (antabuse effect).

19.3 Appendix 3: Drug terminology, common street names and approximate prices of illicit substances

Substance	Measure	Street weight	Approx. cost
Methamphetamine: Chalk, crank, crack, ice, meth and speed, gear.	0.1gram 0.7gram 3.5 grams 7grams 28.3grams	One point Street gram 8 ball (1/8 ounce) 1/4 ball Ounce	\$50-80 \$500 \$800-1000 \$1700 \$5000-7000
Heroin: Big H, horse, hell dust, smack.	0.05 gram 0.1 gram 1 gram 28.3 grams	½ point 1 point Fold/Weight Ounce	\$50 \$60-80 \$500-600 \$10000-12000
Cocaine: Blow, coke, crack, rock, snow.	1gramm 28.3 grams	Ounce	\$350-400 \$8000
Ecstasy: MDMA, eckies, pill, molly	Tablet Tablet	Single Bulk sometimes sheet of 100 tablets	\$10-40
LSD: Acid, trips, Lucy.	Tablet Tablet	Single Bulk sometimes sheet of 100 tablets	\$10-40
Cannabis: Marijuana, weed, cone, choof, grass, pot, gunja, joint, hash	1 gram 28.3grams 454 grams	Bag, foil, stick Ounce, bag Pound (16 ounces)	\$25 - 30 \$250- 300 \$2500-5000
Volatile Substances: Petrol, paint thinner, glue, lighter fluid.	Method of use: huffing, sniffing chroming, bagging		Varies as per chemical used